

Supplementary Information for

Toehold-VISTA: A machine learning approach to decipher programmable RNA sensor-target interactions

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Additional Supplementary Tables:

Supplementary Table 3: Green et al. 2014 mCherry toehold switch benchmark measurements

Supplementary Table 4: Toehold switch and target sequences with experimental measurements

Supplementary Table 5: Calculated biophysical parameters

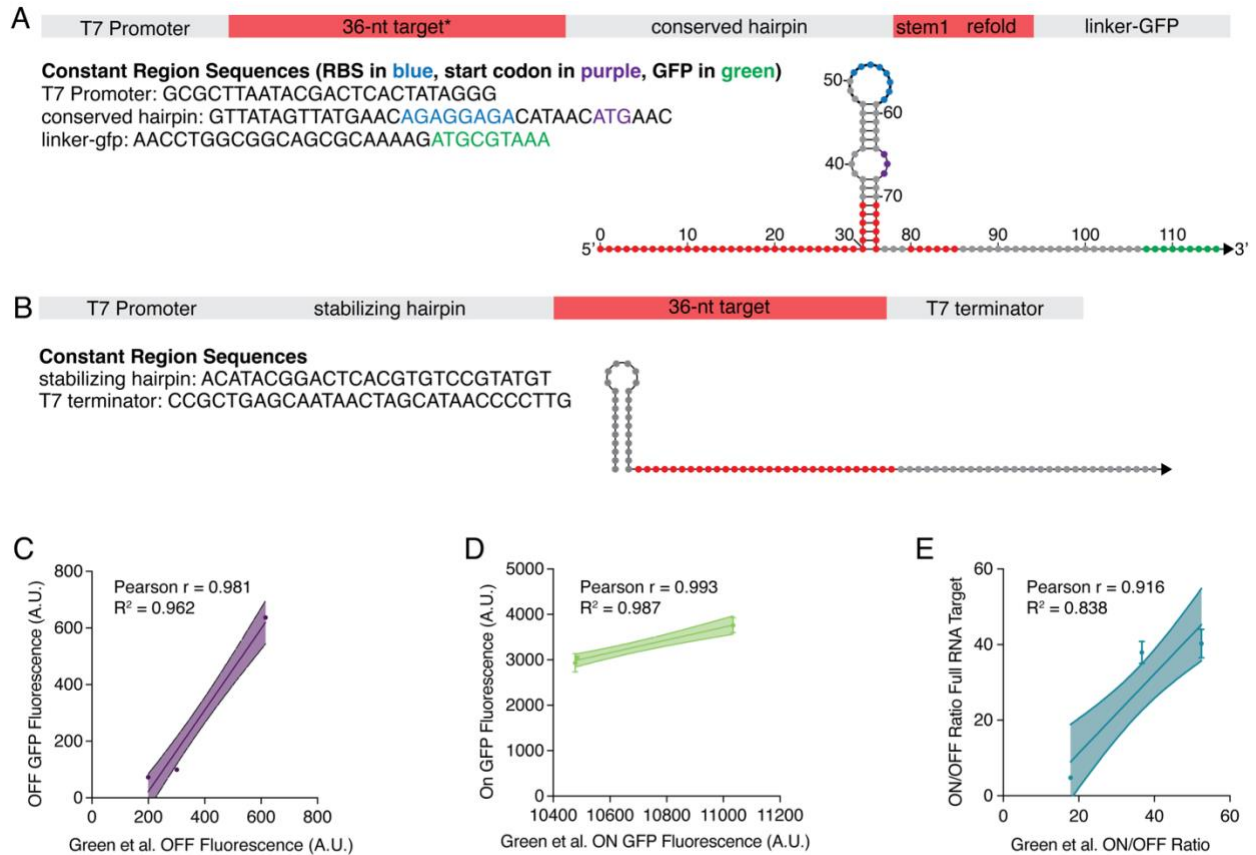
Supplementary Table 6: Pearson r correlations between biophysical parameters and experimental measurements

Supplementary Table 7: Pairwise probabilities for mCherry RNA

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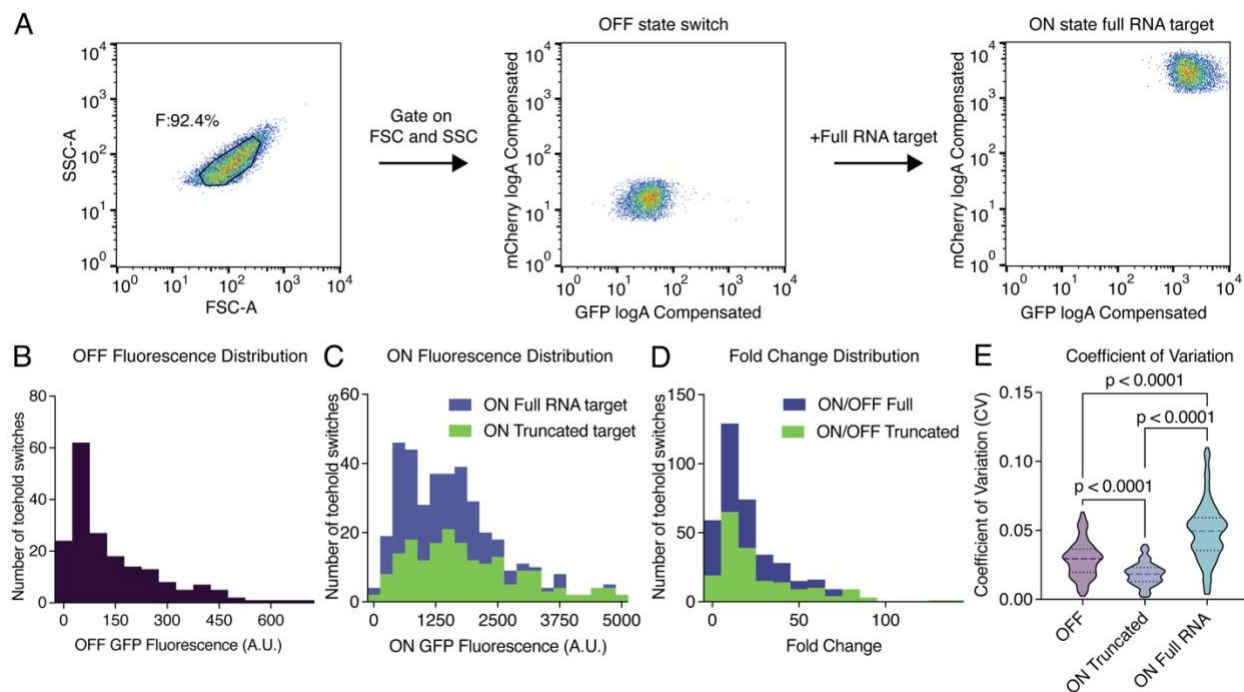
Supplementary Table 9: SARS-CoV-2 N-gene toehold switch and target sequences with experimental flow cytometry measurements

Supplementary Table 10: SARS-CoV-2 N-gene toehold switch cell-free transcription-translation absorbance measurements



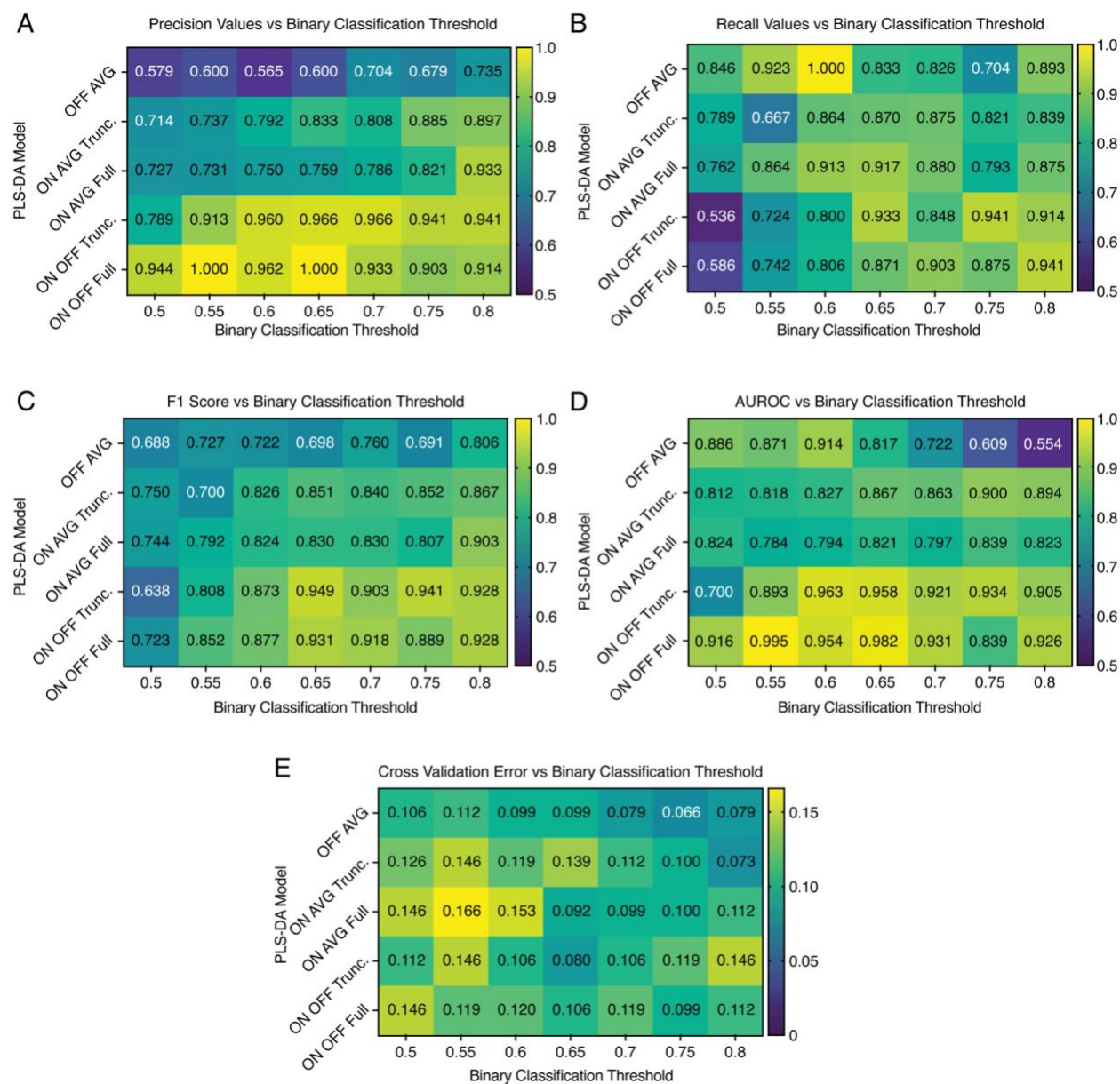
Supplementary Figure 1. Design and validation of toehold switch library. Toehold switch constructs were synthesized as single-stranded DNA and representative switches were verified against previously published switches targeting the *mCherry* mRNA. (A) A schematic of a representative toehold switch construct used for the synthesis of each switch RNA. Toehold switch construct regions include the *promoter* (T7 promoter including GGG region), *36-nucleotide target complement* (individual, toehold-unique), *conserved hairpin* (selected from the tsen2 (1, 2) architecture), *stem1* (complement to the last 6 nucleotides of the trigger binding domain), *refold* (sequesters the bottom part of the toehold switch hairpin after target binding), and *linker-GFP* (21 nucleotide sequence of unstructured amino acids and the first three codons of mut3b-GFP-asv reporter). Further structural details may be found in Supplementary Table 2. (B) Truncated trans RNA target construct design featuring the *promoter* (T7 promoter with GGG region), *stabilizing hairpin* (to stabilize and limit interactions of truncated target with surrounding sequence), *36-nucleotide target* (individual toehold-unique), and *T7 terminator* (first 30 nucleotides of the T7 terminator). The OFF-state GFP fluorescence, ON-state GFP fluorescence, and

fold change of three toehold switches targeting mCherry previously characterized by Green et al. (3) were compared against their three closest respective tiles ordered in this dataset (Supplementary Table 10): (C) OFF-state GFP expression comparison (Pearson $r = 0.981$, $r^2 = 0.962$, error bands indicate 95% confidence interval), (D) the ON-state GFP expression comparison (Pearson $r = 0.993$, $r^2 = 0.987$, error bands indicate 95% confidence interval), and (D) the ON/OFF fold change in GFP expression comparison (Pearson $r = 0.916$, $r^2 = 0.838$, error bands indicate 95% confidence interval).



Supplementary Figure 2. Flow cytometry gating strategy and fluorescence distributions. (A) The gating strategy for IPTG-induced *E. coli* BL21 (DE3) cells is shown. First, cells were gated by their SSC-A and FSC-A distributions. Populations exhibited a unimodal distribution in GFP-logA and mCherry-logA channels. Measurements of OFF-state GFP fluorescence with non-cognate target RNA (B), as well as ON-state GFP fluorescence in presence of truncated or full-length RNA targets (C) and their respective fold changes (D). Coefficient of variation (CV) for fluorescence characterization of each toehold switch variant (E). CV was determined by dividing the standard deviation and the mean of fluorescence measurements for each toehold switch:target variant, derived from each flow cytometry fluorescence characterization described in (B) and Figure 1B. Comparisons were made by one-way ANOVA. For

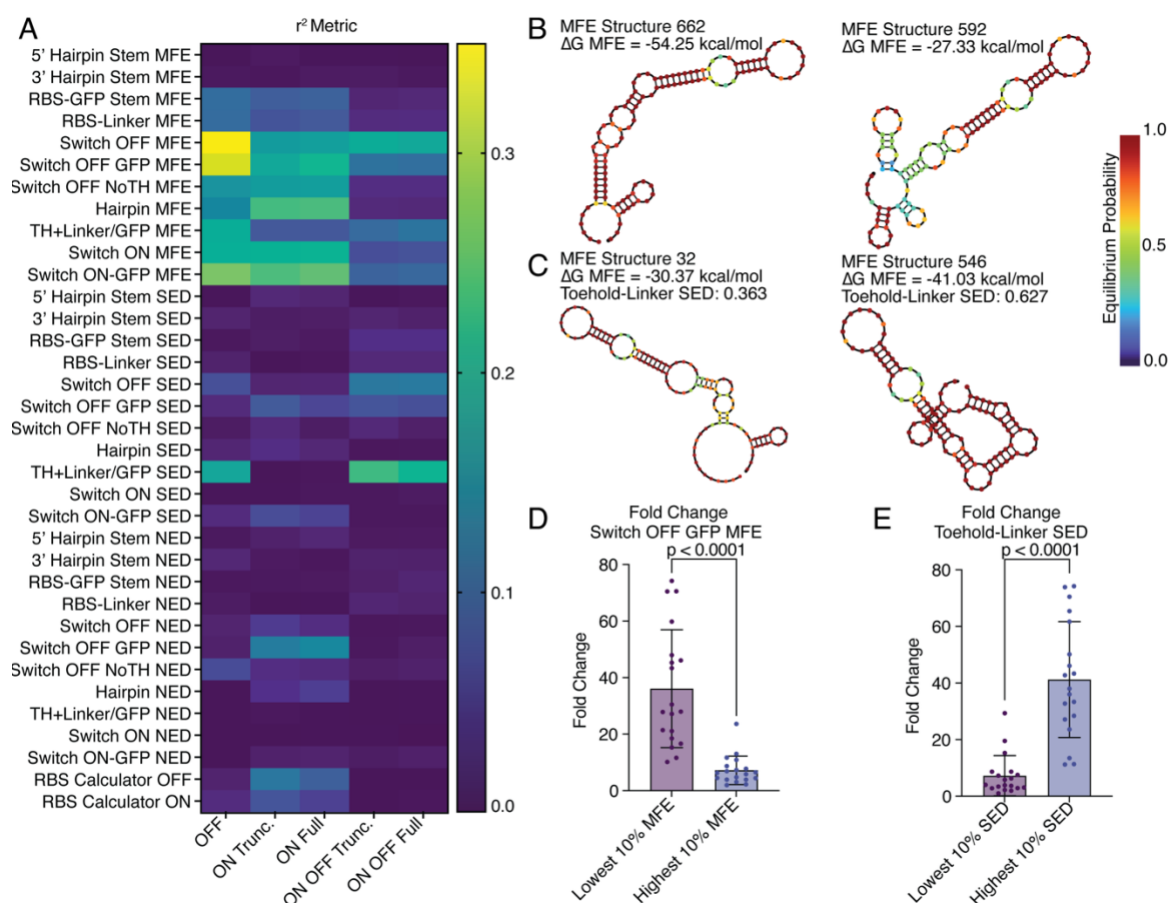
violin plots, the horizontal dashed line represents the median and dotted lines are at 25th and 75th percentiles.



Supplementary Figure 3. Determination of the optimal ON/OFF binary classification threshold.

Precision (predicted positives that are true positives of “high” class), recall (sensitivity or true positive rate), F1 score (harmonic mean of precision and recall), and area under the receiver operator curve (AUROC) were used to determine the optimal cutoff threshold at which to binarize data for each PLS-

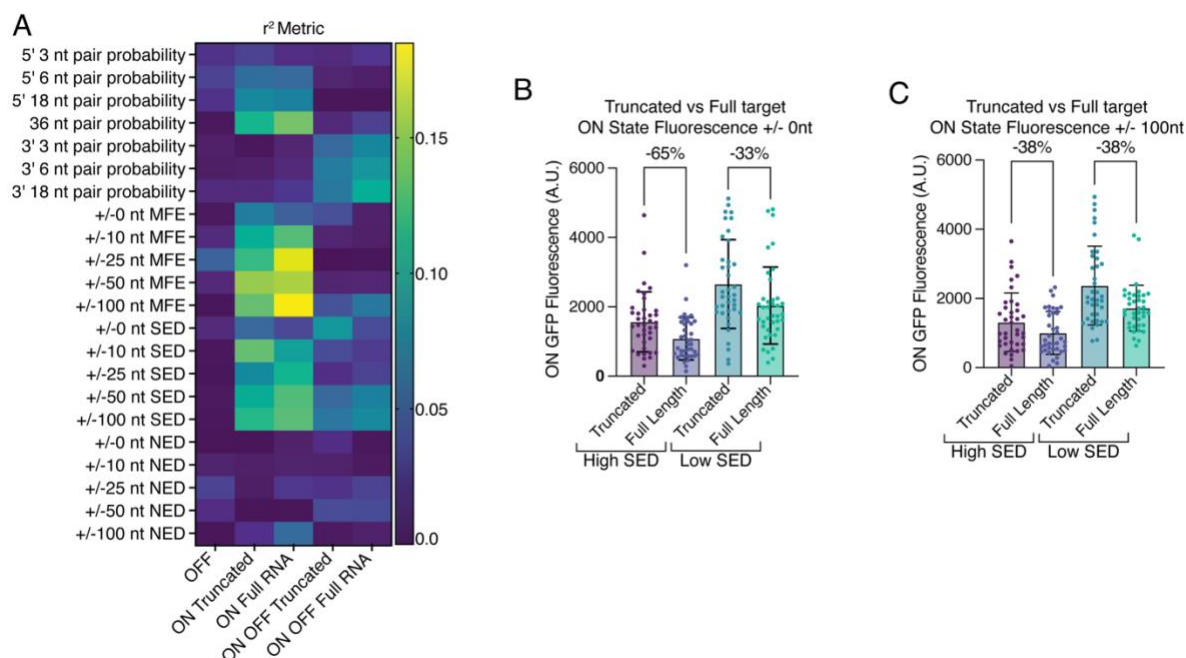
DA model. We trained a PLS-DA model on each of the five empirically determined metrics at seven different binarization threshold and compared the following performance metrics: (A) precision results, (B) recall results, (C) F1 score results, and (D) AUROC results. The final threshold selected for the five classification models was 0.65, specifically for balancing a high precision and recall with the best average AUROC for the truncated and full-RNA target fold change models. We also report the cross-validation error (E), a key metric in the validity of a PLS-DA model, for each binary classification threshold and PLS-DA model trained.



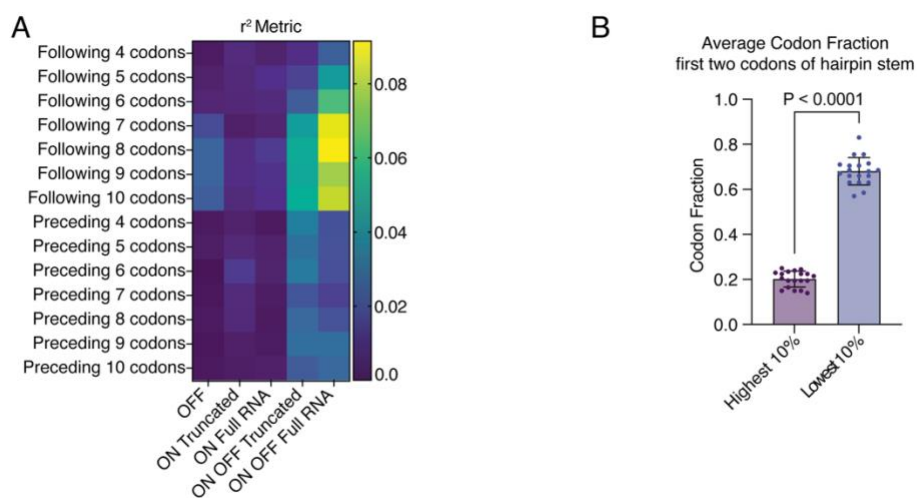
Supplementary Figure 4. Correlations between rational features and the toehold switch dataset. (A)

Squared Pearson correlation coefficients ($\max(r^2) = 0.35$) from Figure 2A. (B) Representative NUPACK-determined minimum free energy (MFE) structures for the lowest (switch 662) and highest (switch 592) MFE values for the Switch-OFF MFE metric. (C) Representative NUPACK-determined MFE structures for the lowest (switch 32) and highest (switch 546) toehold-linker/GFP ideal ensemble defect (IED). In each case, the region preceding the hairpin stem should be totally unbound, as shown by the

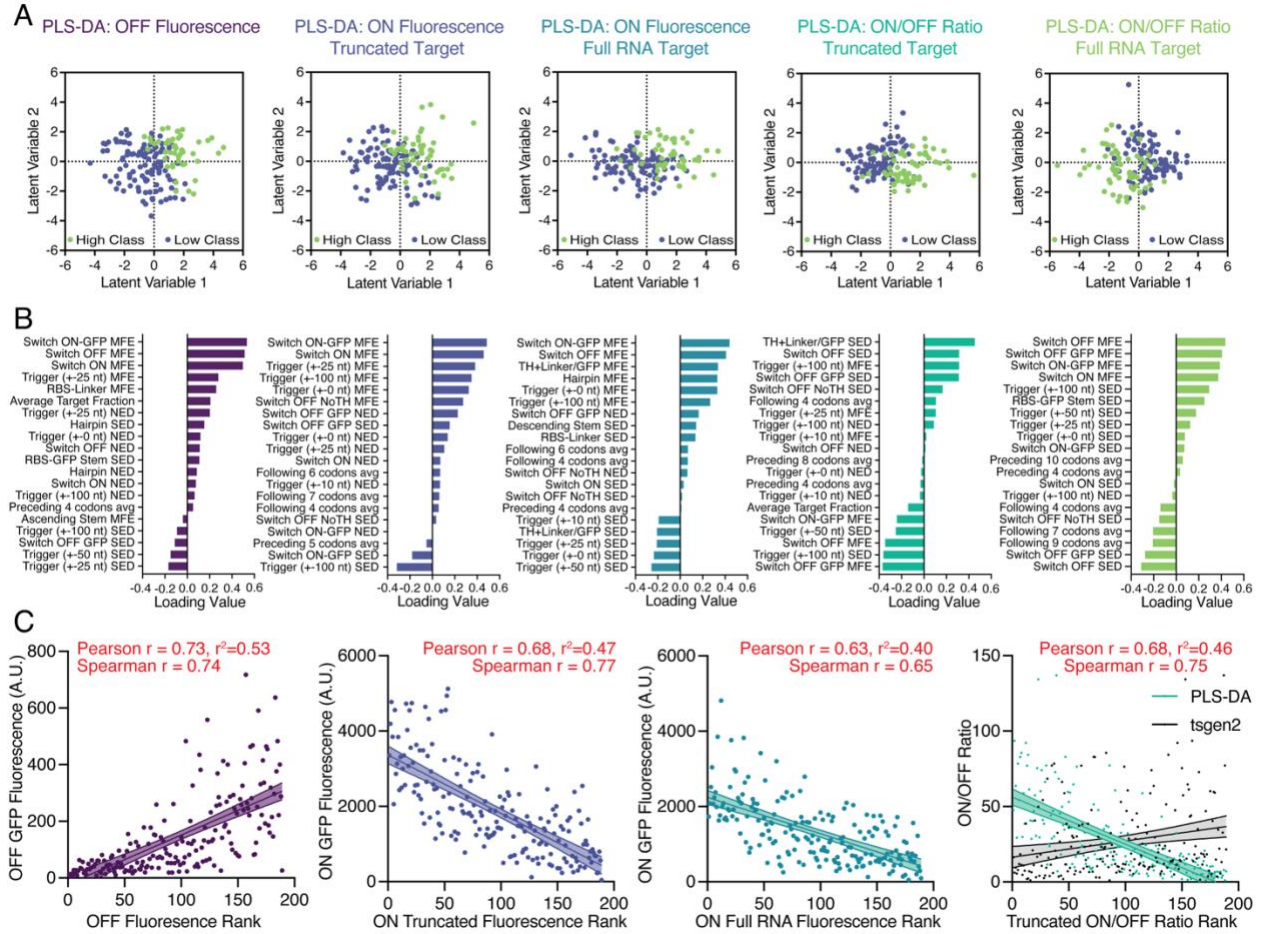
ideal structure in Supplementary Figure 1A. (D) Comparison of the fold change for full RNA target when stratified by the lowest and highest 10% MFE for the “Switch OFF GFP” metric ($n=19$, $p < 0.0001$ two-tailed t test). (E) Comparison of full RNA target fold change when stratified for lowest and highest 10% IED for the “Toehold linker/GFP” metric. For all bar plots, horizontal line indicates the mean, and whiskers indicate the standard deviation (SD).



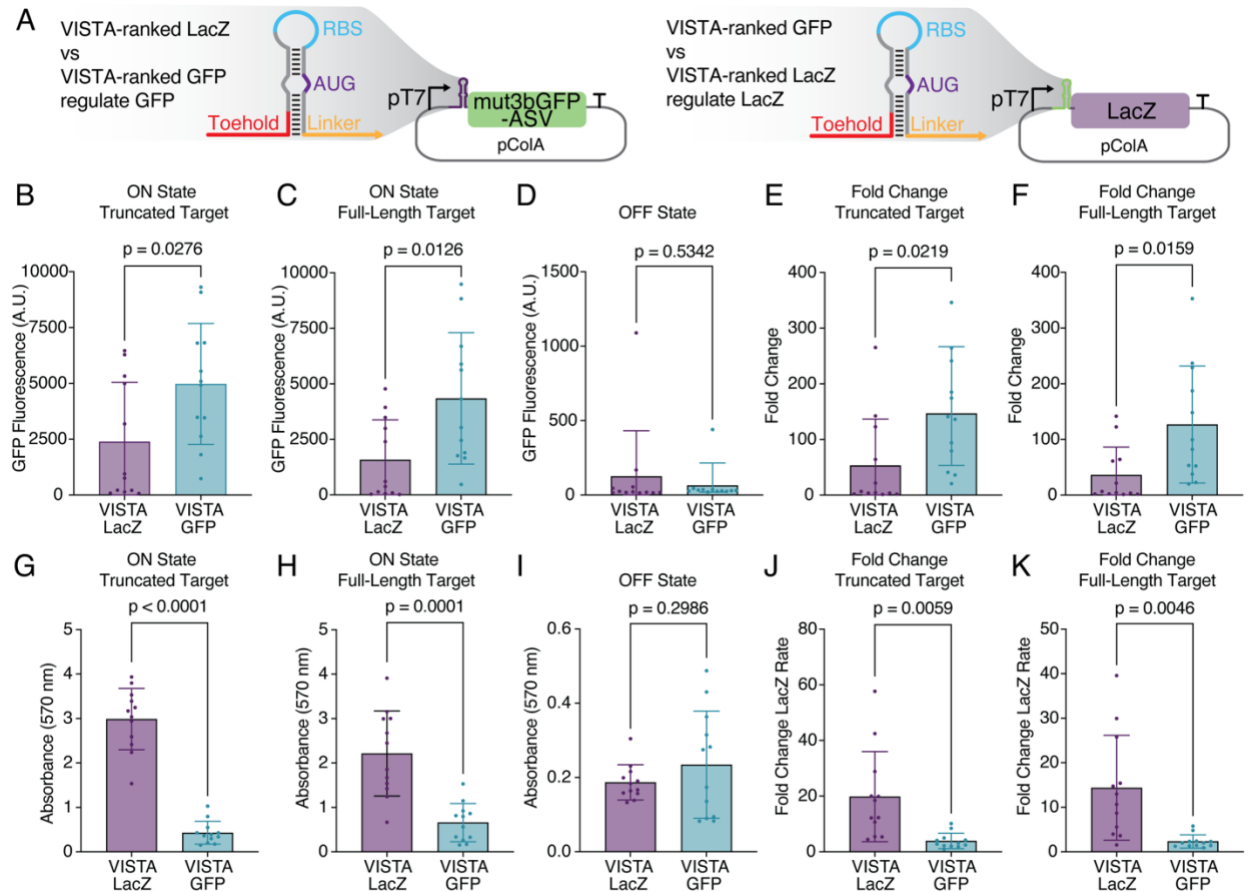
Supplementary Figure 5. Correlations between rational features of the full RNA target and the toehold switch dataset. (A) Squared Pearson correlation coefficients ($\max(r^2) = 0.18$) of correlations from Figure 3B. (B) Truncated and full-length ON-state GFP fluorescence were stratified for the highest and lowest 20% ideal ensemble defect (IED) metric for 0 nucleotide flanking region of the full-length target ($n = 38$). (C) Truncated and full-length ON-state GFP fluorescence were stratified for the highest and lowest 20% IED metric for 100-nucleotide flanking region of the full-length target ($n=38$). For all bar plots, horizontal line indicates the mean, and whiskers indicate the standard deviation (SD).



Supplementary Figure 6. Correlations between codon fraction around full RNA target and toehold switch dataset. (A) Squared Pearson correlation coefficients ($\max(r^2) = 0.09$) from Figure 3B. (B) Lowest and highest 10% of average codon fractions in the first two codons at the base of toehold switch hairpin stem ($n=19$, $p < 0.0001$, two-tailed t test). For all bar plots, horizontal line indicates the mean, and whiskers indicate the standard deviation (SD).



Supplementary Figure 7. Feature contributions and validation of PLS-DA models. (A) Latent variable scores plot for each of the five independently trained PLS-DA models. (B) PLS-DA loading plots showing feature contributions to the first latent variable for each performance model. Features with the highest absolute loadings indicate the most influential rational parameter in classifying toehold switch performance. (C) Experimental values plotted against PLS-DA algorithm rank for OFF-state GFP fluorescence, ON-state truncated GFP fluorescence, ON-state full RNA target GFP fluorescence, and truncated ON/OFF fold-change ratio. Linear regression is plotted with 95% confidence interval.



Supplementary Figure 8. Contextual effects of output genes on toehold switch function. (A) Schematic comparing GFP and LacZ reporter configurations used to evaluate how VISTA-ranked toehold switches designed for LacZ and GFP reporters function when the output gene is exchanged. (A-C) ON-state GFP fluorescence for truncated (B) and full-length (C) RNA targets and (D) OFF-state GFP fluorescence in BL 21 Star DE3 bacteria. (E-F) Fold change values for truncated (E) and full-length (F) targets with GFP reporter. (G-I) Endpoint ON-state LacZ absorbance measurements at 2.5 hours for truncated (G) and full-length (H) targets and (I) OFF-state LacZ absorbance in cell-free transcription-translation reactions. (J-K) Fold change in LacZ reaction rates at 60 minutes for truncated (J) and full-length (K) targets. Comparisons made with two-tailed t tests. For all bar plots, horizontal bar represents the mean and whiskers the SD.

Primer Name	Sequence	Primer Function
Switch_BB_fwd	AACCTGGCGGCAGCGCAAAAGATGCGTAAAGGA GAAGAACTTTTCACT	Used to linearize ColA plasmid backbone
Switch_BB_rev	TGTTGGGGTTCTCTTAGCTTTGTTTCGCCGCATAA GGGAGAGCGTCGAGATC	Used to linearize ColA plasmid backbone
Trigger_BB_fwd	CCGCTGAGCAATAACTAGCATAACC	Used to linearize ColE1 plasmid backbone
Trigger_BB_rev	GAGCTATATCGCGAACCACTGGCAGACTACCGAG ATCTCGATCCTCTACGC	Used to linearize ColE1 plasmid backbone
Switch_insert_fwd	CGGCGAAACAAAGCTAAGAGAACCCCAACAGCG CTAATACGACTCACTATAGGG	Used to amplify toehold switch inserts for ColA- GFP backbone
Switch_insert_rev	TTTACGCATCTTTTGGCGTGCCGCCAGGTT	Used to amplify toehold switch inserts for ColA- GFP backbone
Trigger_insert_fwd	GTAGTCTGCCAGTGTTTCGCGATATAGCTCGCGCT AATACGACTCACTATAGGG	Used to amplify 36 nucleotide truncated targets for ColE1 backbone
Trigger_insert_rev	CAAGGGGTTATGCTAGTTATTGCTCAGCGG	Used to amplify 36 nucleotide truncated targets for ColE1 backbone
GFPmut3b-ASV	ATGCGTAAAGGAGAAGAACTTTTCACTGGAGTTG TCCCAATTCTTGTTGAATTAGATGGTGATGTTAAT GGGCACAAATTTTCTGTCACTGGAGAGGGTGAAG GTGATGCAACATACGGAAAACCTTACCCTTAAATT TATTTGCACTACTGGAAAACCTGTTCCGTGGC CAACACTTGTCACACTTTTCGGTTATGGTGTTCAA TGCTTTGCGAGATACCCAGATCACATGAAACAGC ATGACTTTTTCAAGAGTGCCATGCCCCAAGGTTA CGTACAGGAAAGAAGTATATTTTTCAAAGATGAC GGGAACTACAAGACACGTGCTGAAGTCAAGTTTG AAGGTGATACCCTTGTTAATAGAATCGAGTAAA AGGTATTGATTTTAAAGAAGATGGAAACATTCTT GGACACAAATTGGAATACAATACTACACACA ATGTATACATCATGGCAGACAAACAAAAGAATGG AATCAAAGTTAACTTCAAAATTAGACACAACATT	Used as toehold switch reporter

	GAAGATGGAAGCGTTCAACTAGCAGACCATTATC AACAAAATACTCCGATTGGCGATGGCCCTGTCCT TTTACCAGACAACCATTACCTGTCCACACAATCTG CCCTTTCGAAAGATCCCAACGAAAAGAGAGACCA CATGGTCCTTCTTGAGTTTGTAAACCGCTGCTGGGA TTACACATGGCATGGATGAACTATACAAAAGGCC TGCAGCAAACGACGAAAACCTACGCTGCATCAGTT TAATAA	
mCherry	ATGGTGAGCAAGGGCGAGGAGGATAACATGGCC ATCATCAAGGAGTTCATGCGCTTCAAGGTTTACA TGGAGGGCTCCGTGAACGGCCACGAGTTCGAGAT CGAGGGCGAGGGCGAGGGCCGCCCTACGAGGG CACCCAGACCGCCAAGCTGAAGGTGACCAAGGGT GGCCCCCTGCCCTTCGCCTGGGACATCCTGTCCCC TCAGTTCATGTACGGCTCCAAGGCCTACGTGAAG CACCCCGCCGACATCCCCGACTACTTGAAGCTGTC CTTCCCCGAGGGCTTCAAGTGGGAGCGCGTGATG AACTTCGAGGACGGCGGCGTGGTGACCGTGACCC AGGACTCCTCCCTGCAAGACGGCGAGTTCATCTA CAAGGTGAAGCTGCGCGGCACCAACTTCCCCTCC GACGGCCCCGTAATGCAGAAGAAGACTATGGGCT GGGAGGCCTCCTCCGAGCGGATGTACCCCGAGGA CGGCGCGCTGAAGGGCGAGATCAAGCAGAGGCT GAAGCTGAAGGACGGCGGCCACTACGACGCTGA GGTCAAGACCACCTACAAGGCCAAGAAGCCCGTG CAACTGCCCGGCGCGTACAACGTCAACATCAAGT TGGACATCACCTCCCACAACGAGGACTACACCAT CGTGGAACAGTACGAACGCGCCGAGGGCCGCCAC TCCACCGGCGGCATGGACGAGCTGTACAAGTAA	Used as full RNA target
LacZ	ATGACCATGATTACGGATTCACTGGCCGTCGTTTT ACAACGTCGTGACTGGGAAAACCCTGGCGTTACC CAACTTAATCGCCTTGACGACATCCCCCTTTTCGC CAGCTGGCGTAATAGCGAAGAGGCCCGCACCGAT CGCCCTTCCCAACAGTTGCGCAGCCTGAATGGCG AATGGCGCTTTGCCTGGTTTCCGGCACCCAGAAGC GGTGCCGAAAAGCTGGCTGGAGTGCGATCTTCCT GAGGCCGATACTGTCGTGTCCTCCCTCAAACCTGGC AGATGCACGGTTACGATGCGCCCATCTACACCAA CGTGACCTATCCCATTACGGTCAATCCGCGGTTTG TTCCACGGAGAATCCGACGGGTTGTTACTCGCTC ACATTTAATGTTGATGAAAGCTGGCTACAGGAAG GCCAGACGCGAATTATTTTTGATGGCGTTAACTCG GCGTTTCATCTGTGGTGCAACGGGCGCTGGGTCTG GTTACGGCCAGGACAGTCGTTTGCCGTCTGAATTT GACCTGAGCGCATTTTTACGCGCCGGAGAAAACC GCCTCGCGGTGATGGTGCTGCGCTGGAGTGACGG CAGTTATCTGGAAGATCAGGATATGTGGCGGATG	Used as toehold switch reporter in cell-free transcription-translation reactions

	AGCGGCATTTTCCGTGACGTCTCGTTGCTGCATAA ACCGACTACACAAATCAGCGATTTCCATGTTGCC ACTCGCTTTAATGATGATTTTCAGCCGCGCTGTACT GGAGGCTGAAGTTCAGATGTGCGGCGAGTTGCGT GACTACCTACGGGTAACAGTTTCTTTATGGCAGG GTGAAACGCAGGTCGCCAGCGGCACCGCGCCTTT CGGCGGTGAAATTATCGATGAGCGTGGTGGTTAT GCCGATCGCGTCACACTACGTCTGAACGTCGAAA ACCCGAACTGTGGAGCGCCGAAATCCCGAATCT CTATCGTGCGGTGGTTGAACTGCACACCGCCGAC GGCACGCTGATTGAAGCAGAAGCCTGCGATGTCG GTTTCCGCGAGGTGCGGATTGAAAATGGTCTGCT GCTGCTGAACGGCAAGCCGTTGCTGATTTCGAGGC GTTAACCGTCACGAGCATCATCCTCTGCATGGTCA GGTCATGGATGAGCAGACGATGGTGCAAGGATATC CTGCTGATGAAGCAGAACAACTTTAACGCCGTGC GCTGTTTCGATTATCCGAACCATCCGCTGTGGTAC ACGCTGTGCGACCGCTACGGCCTGTATGTGGTGG ATGAAGCCAATATTGAAACCCACGGCATGGTGCC AATGAATCGTCTGACCGATGATCCGCGCTGGCTA CCGGCGATGAGCGAACGCGTAACGCGAATGGTGC AGCGCGATCGTAATCACCCGAGTGTGATCATCTG GTCGCTGGGGAATGAATCAGGCCACGGCGCTAAT CACGACGCGCTGTATCGCTGGATCAAATCTGTCG ATCCTTCCCGCCCGGTGCAGTATGAAGGCGGCGG AGCCGACACCACGGCCACCGATATTATTTGCCCCG ATGTACGCGCGCGTGGATGAAGACCAGCCCTTCC CGGCTGTGCCGAAATGGTCCATCAAAAAATGGCT TTCGCTACCTGGAGAGACGCGCCCGCTGATCCTTT GCGAATACGCCCACGCGATGGGTAACAGTCTTGG CGGTTTCGCTAAATACTGGCAGGCGTTTCGTCAGT ATCCCCGTTTACAGGGCGGCTTCGCTCTGGGACTG GGTGGATCAGTCGCTGATTAAATATGATGAAAAC GGCAACCCGTGGTCGGCTTACGGCGGTGATTTTG GCGATACGCCGAACGATCGCCAGTTCTGTATGAA CGGTCTGGTCTTTGCCGACCGCACGCCGCATCCAG CGCTGACGGAAGCAAAACACCAGCAGCAGTTTTT CCAGTTCCGTTTATCCGGGCAAACCATCGAAGTG ACCAGCGAATACCTGTTCCGTCATAGCGATAACG AGTCCTGCACTGGATGGTGGCGCTGGATGGTAA GCCGCTGGCAAGCGGTGAAGTGCCTCTGGATGTC GCTCCACAAGGTAAACAGTTGATTGAACTGCCTG AACTACCGCAGCCGGAGAGCGCCGGGCAACTCTG GCTCACAGTACGCGTAGTGCAACCGAACGCGACC GCATGGTCAGAAGCCGGGCACATCAGCGCCTGGC AGCAGTGGCGTCTGGCGGAAAACCTCAGTGTGAC GCTCCCCGCCGCTCCACGCCATCCCGCATCTGA CCACCAGCGAAATGGATTTTTGCATCGAGCTGGG TAATAAGCGTTGGCAATTTAACCGCCAGTCAGGC	
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	TTTCTTTTCACAGATGTGGATTGGCGATAAAAAAC AACTGCTGACGCCGCTGCGCGATCAGTTCACCCG TGCACCGCTGGATAACGACATTGGCGTAAGTGAA GCGACCCGCATTGACCCTAACGCCTGGGTCTGAAC GCTGGAAGGCGGCGGGCCATTACCAGGCCGAAGC AGCGTTGTTGCAGTGCACGGCAGATACACTTGCT GATGCGGTGCTGATTACGACCGCTCACGCGTGGC AGCATCAGGGGAAAACCTTATTTATCAGCCGGAA AACCTACCGGATTGATGGTAGTGGTCAAATGGCG ATTACCGTTGATGTTGAAGTGGCGAGCGATACAC CGCATCCGGCGCGGATTGGCCTGAACTGCCAGCT GGCGCAGGTAGCAGAGCGGGTAAACTGGCTCGG ATTAGGGCCGCAAGAAAACCTATCCCGACCGCCTT ACTGCCGCTGTTTTGACCGCTGGGATCTGCCATT GTCAGACATGTATACCCCGTACGTCTTCCCGAGC GAAAACGGTCTGCGCTGCGGGACGCGCGAATTGA ATTATGGCCACACCCAGTGGCGCGGGCGACTTCCA GTTCAACATCAGCCGCTACAGTCAACAGCAACTG ATGGAAACCAGCCATCGCCATCTGCTGCACGCGG AAGAAGGCACATGGCTGAATATCGACGGTTTCCA TATGGGGATTGGTGGCGACGACTCCTGGAGCCCG TCAGTATCGGCGGAATTCCAGCTGAGCGCCGGTC GCTACCATTACCAGTTGGTCTGGTGTCAAAAA	
SARS-CoV-2 Nucleocapsid Protein	TATACCATGGGCAGCAGCATGTCTGATAATGGAC CCCAAAATCAGCGAAATGCACCCCGCATTACGTT TGGTGGACCCTCAGATTCAACTGGCAGTAACCAG AATGGAGAACGCAGTGGGGCGCGATCAAAACAA CGTCGGCCCCAAGGTTTACCCAATAATACTGCGT CTTGTTTACCGCTCTCACTCAACATGGCAAGGA AGACCTTAAATTCCCTCGAGGACAAGGCGTTCCA ATTAACACCAATAGCAGTCCAGATGACCAAATTG GCTACTACCGAAGAGCTACCAGACGAATTTCGTGG TGGTGACGGTAAAATGAAAGATCTCAGTCCAAGA TGGTATTTCTACTACCTAGGAACTGGGCCAGAAG CTGGACTTCCCTATGGTGCTAACAAAGACGGCAT CATATGGGTTGCAACTGAGGGAGCCTTGAATACA CCAAAAGATCACATTGGCACCCGCAATCCTGCTA ACAATGCTGCAATCGTGCTACAACCTTCCTCAAGG AACAACATTGCCAAAAGGCTTCTACGCAGAAGGG AGCAGAGGCGGCAGTCAAGCCTCTTCTCGTTCCTC ATCACGTAGTCGCAACAGTTCAAGAAATTCAACT CCAGGCAGCAGTAGGGGAACCTTCTCCTGCTAGAA TGGCTGGCAATGGCGGTGATGCTGCTCTTGCTTTG CTGCTGCTTGACAGATTGAACCAGCTTGAGAGCA AAATGTCTGGTAAAGGCCAACAACAACAAGGCCA AACTGTCACTAAGAAATCTGCTGCTGAGGCTTCT AAGAAGCCTCGGCAAAAACGTACTGCCACTAAAG CATACAATGTAACACAAGCTTTCGGCAGACGTGG TCCAGAACAAACCAAGGAAATTTTGGGGACCAG	Used in Figure 5 for full- length RNA target

	GAAC TAATCAGACAAGGAACTGATTACAAACATT GGCCGCAAATTGCACAATTTGCCCCAGCGCTTC AGCGTTCTTCGGAATGTCGCGCATTGGCATGGAA GTCACACCTTCGGGAACGTGGTTGACCTACACAG GTGCCATCAAATTGGATGACAAAGATCCAAATTT CAAAGATCAAGTCATTTTGCTGAATAAGCATATT GACGCATACAAAACATTCCCACCAACAGAGCCTA AAAAGGACAAAAAGAAGAAGGCTGATGAACTC AAGCCTTACCGCAGAGACAGAAGAAACAGCAAA CTGTGACTCTTCTTCTGCTGCAGATTTGGATGAT TTCTCCAAACAATTGCAACAATCCATGAGCAGTG CTGACTCAACTCAGGCCTAA	
T7 terminator	ccgctgAgcAAtAActAgcAtAAcccttgAGATAACAGAT ACttcgGtAtctgttAtctgttTTTTTTcAACAGATAGCCGC GttcgCGCGGcAtctgttTTTTTT	

Supplementary Table 1. Sequences for plasmid construction. A list of all primers used to construct toehold switch and truncated target plasmids including their name, sequence, and primary function. Also included are reporter protein sequences and the T7 terminator variant used in this study.

Rational Feature Name	Sequence Region (nucleotide range)	Run-length-encoded (RLE) dot-parens-plus And DU+ Notation	Description
5' hairpin stem	30:48	‘.18’ or U18	Switch hairpin stem Ascending sequence
3' hairpin stem	58:77	‘.18’ or U18	Switch hairpin stem descending sequence
RBS-GFP	48:116	‘.20 (6 .6)6 .30’ or U20 D6 U6 U30	RBS sequence to start of GFP
RBS-Linker	48:107	‘.20 (6 .6)6 .21’ or U20 D6 U6 U21	RBS sequence to end of linker
Switch OFF	0:77	‘.30 (9 .3 (6 .11)6 .3)9’ or U30 D9 (U3 D6 U11 U3)	OFF conformation of toehold switch

Switch OFF-GFP	0:116	‘.30 (9 .3 (6 .11)6 .3)9 .30’ or U30 D9 (U3 D6 U11 U3) U30	OFF conformation of toehold switch to GFP start
Switch OFF No Toehold	60:116	‘(9 .3 (6 .11)6 .3)9 .30’ or D9 (U3 D6 U11 U3) U30	OFF conformation of toehold switch without toehold
Hairpin	30:77	‘(9 .3 (6 .11)6 .3)9’ or D9 (U3 D6 U11 U3)	Hairpin stem of the toehold switch
Toehold + Linker/GFP	0:30 + 77:116	‘.30 + .30’ or U30 + U30	No hairpin stem; toehold and linker-GFP sequence
Switch ON	0:76	‘(36 +)36 .41’ or D36 + U41	ON conformation of toehold switch to linker
Switch ON-GFP	0:116	‘(36 +)36 .6 (6 .11)6 .3 (6 .6)6 .30’ or D36 + U6 D6 U11 U3 D6 U6 U30	ON conformation of toehold switch to GFP

Supplementary Table 2. Rational feature sub-sequence regions and description. Individual sub-regions of the toehold switch from which the 33 rational features were calculated in Figure 2A using NUPACK And ViennaRNA. Sequence nucleotide ranges are reported from the 5’ most position (index 0) to the 3’ most position (index 116); see Supplementary Fig. 1.